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Impacts of PEMF Applications on Photoplethysmography Responses

AUTHORS

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Introduction

Understanding the impact of Pulsed Electromagnetic Fields (PEMF) on blood flow is crucial for advancements in healthcare. This study investigated how different PEMF configurations affect peripheral blood flow, measured using Photoplethysmography (PPG). We explored four PEMF settings varying in frequency and pulse width while maintaining a constant current. We then analyzed changes in key PPG signal characteristics, including cycle time, peak amplitudes, notch amplitude, and the power spectral density (PSD). By comparing these measures across PEMF configurations and a control group (no PEMF), we aimed to understand how PEMF influences blood flow and its potential for future medical applications.

Objectives

This study aimed to investigate the effects of various Pulsed Electromagnetic Field (PEMF) configurations on peripheral blood flow as measured by Photoplethysmography (PPG).

01 Evaluate the impact of different PEMF configurations on PPG signal characteristics.

02 Assess changes in cycle time, peak amplitudes, and notch amplitude compared to No-PEMF.

03 Investigate the frequency and current dependence of observed changes.

Methodology

Data Collection

Fourteen healthy subjects were involved in our study, which received institutional ethics approval. A circular PEMF coil, 8 cm in diameter, was placed on the forearm. A PEMF probe was placed on the tip of the index finger to observe the effects of PEMF on blood flow. Four categories of PEMF were applied. PEMF pulses were administered to the healthy subjects for 10 minutes following a one-minute no-PEMF condition. Data were collected across several metrics, including timestamp, device ID, battery status, PPG signal, heart rate, spO2, ADC readings from red and infrared sensors, spO2 status and perfusion index. All data were collected and stored in Excel sheets for further evaluation.

4 different setups with variations in frequency (Hz), pulse width (μsec), and current (A)

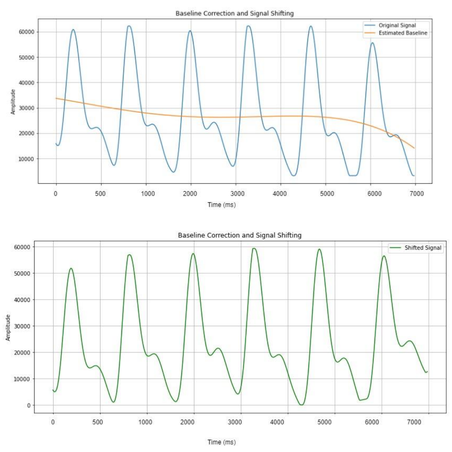
PEMF Setup	Frequency (Hz)	Pulse width (μsec)	Current (A)
PEMF-1	10	100	3.5
PEMF-2	10	500	3.5
PEMF-3	70	100	3.5
PEMF-4	70	500	3.5



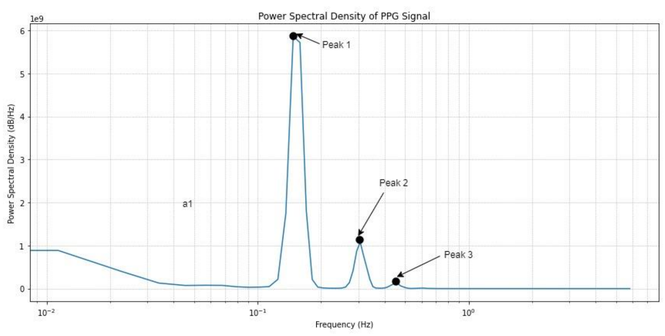
Data Processing

Power Spectral Density as well as Time domain were analyzed from all the recorded data in terms of their four setups of PEMF and no PEMF conditions. Using the data, we extracted various key features both in the time and frequency domains. Various comparative analyses have been conducted to visualize and identify any discernible impact due to the application of PEMF. All the calculated values were compared among the no PEMF PPG signals and the four setups of PEMF applications. We used MATLAB, Python, and Excel to perform all these tasks.

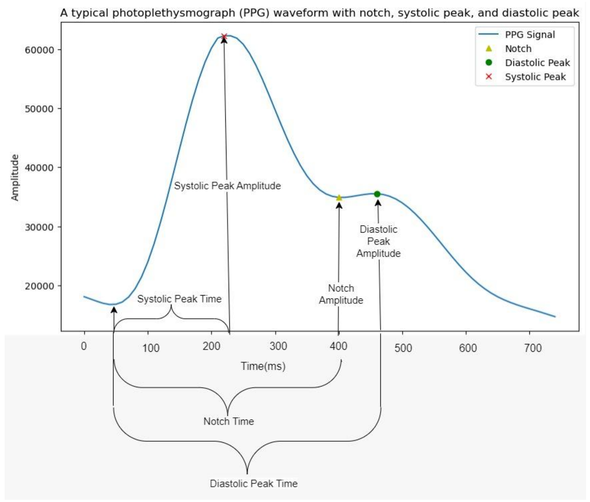
Baseline correction and signal shifting



Power Spectral Density



Time Domain Features



Systolic Peak

Diastolic Peak

Height of Notch

Systolic Peak Time

Diastolic Peak Time

Height of Notch Time

Cycle Time

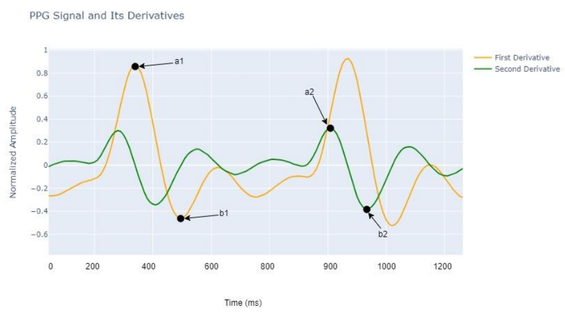
Change with respect to No PEMF

PEMF Setup	Cycle Time	Systolic Peak	Diastolic Peak	Height of Notch
(i) PEMF-1	**8.57%**	0.29%	3.83%	4.16%
(ii) PEMF-2	1.14%	0.75%	4.45%	**5.67%**
(iii) PEMF-3	2.40%	1.59%	1.73%	1.49%
(iv) PEMF-4	0.61%	**2.80%**	**4.77%**	2.68%

Features derived from the first and second derivative

The first maximum peak from the first derivative of the PPG waveform (a1)
The time interval from the foot of the PPG waveform to the time at which a1 occurred (ta1)
The first minimum peak from the first derivative of the PPG waveform after a1 (b1)
The time interval from the foot of the PPG waveform to the time at which b1 occurred (tb1)

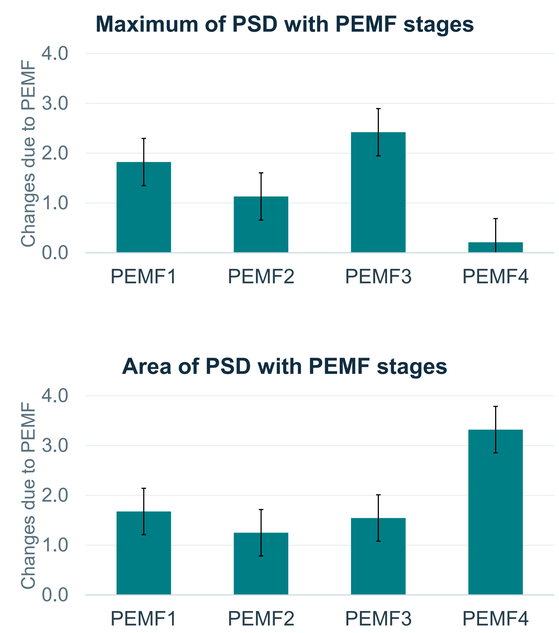
The first maximum peak from the second derivative of the PPG waveform after a1 (a2)
The time interval from the foot of the PPG waveform to the time at which a2 occurred (ta2)
The first minimum peak from the second derivative of the PPG waveform after a2 (b2)
The time interval from the foot of the PPG waveform to the time at which b2 occurred (tb2)

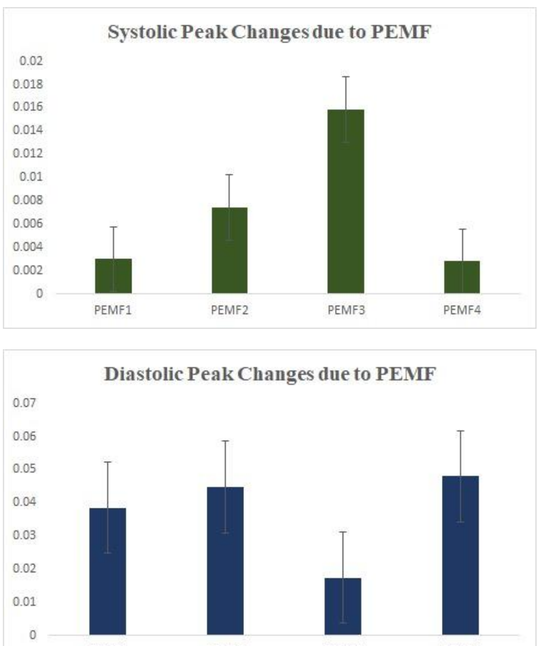


Change with respect to No PEMF

PEMF Setup	a1	a2	b1	b2
(i) PEMF-1	**2.25%**	17.69%	**22.86%**	20.61%
(ii) PEMF-2	0.60%	6.38%	4.02%	22.30%
(iii) PEMF-3	1.16%	3.72%	4.59%	7.16%
(iv) PEMF-4	0.26%	**21.36%**	17.23%	**28.23%**

Results





Conclusion

Our analysis confirms that PEMF can significantly alter arterial blood flow, showcasing its potential to impact related biophysical parameters. These findings suggest that PEMF could be utilized therapeutically to modulate blood flow in clinical settings. Its applications could play a significant role in improving vascular health and treating conditions associated with impaired blood circulation.

References

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